
Class A

- Definition: Class A networks are designed for large-scale organizations requiring millions of IP addresses.
- Range: 1.0.0.0 - 126.255.255.255
- Subnet Mask: 255.0.0.0 (8-bit network, 24-bit host)
- Usage: Government agencies, major corporations, and ISPs.
- Number of Hosts per Network: Over 16 million
- Example: A company like Google or Amazon may use a Class A network for extensive infrastructure.

Class B

- Definition: Class B networks serve medium-sized businesses, universities, and organizations requiring thousands of IP addresses.
- Range: 128.0.0.0 - 191.255.255.255
- Subnet Mask: 255.255.0.0 (16-bit network, 16-bit host)
- Usage: Universities, large enterprises, hospitals, and government agencies.
- Number of Hosts per Network: Up to 65,534
- Example: Universities with multiple campuses use Class B networks to manage various sub-networks.

Class C

- Definition: Class C networks are best suited for small businesses, offices, and home networks where fewer addresses are needed.
- Range: 192.0.0.0 - 223.255.255.255
- Subnet Mask: 255.255.255.0 (24-bit network, 8-bit host)
- Usage: Used for LANs (Local Area Networks), home Wi-Fi, and small corporate networks.
- Number of Hosts per Network: 254
- Example: Home Wi-Fi networks and small office setups typically use Class C addresses.

Class D (Multicast)

- Definition: Class D is not used for traditional network allocation. It is specifically reserved for multicast traffic—where one sender distributes data to multiple recipients simultaneously.
- Range: 224.0.0.0 - 239.255.255.255
- Subnet Mask: None (used for multicast communication)
- Usage: Streaming services, real-time conferencing, and gaming.
- Number of Hosts per Network: Varies; used for group communication rather than direct addressing.
- Example: Online gaming or video conferencing software relies on multicast addresses to distribute data efficiently.

Class E (Experimental)

- Definition: Class E is reserved for future use and research. It is not used for standard networking.
- Range: 240.0.0.0 - 255.255.255.255
- Usage: Reserved for research and testing purposes.